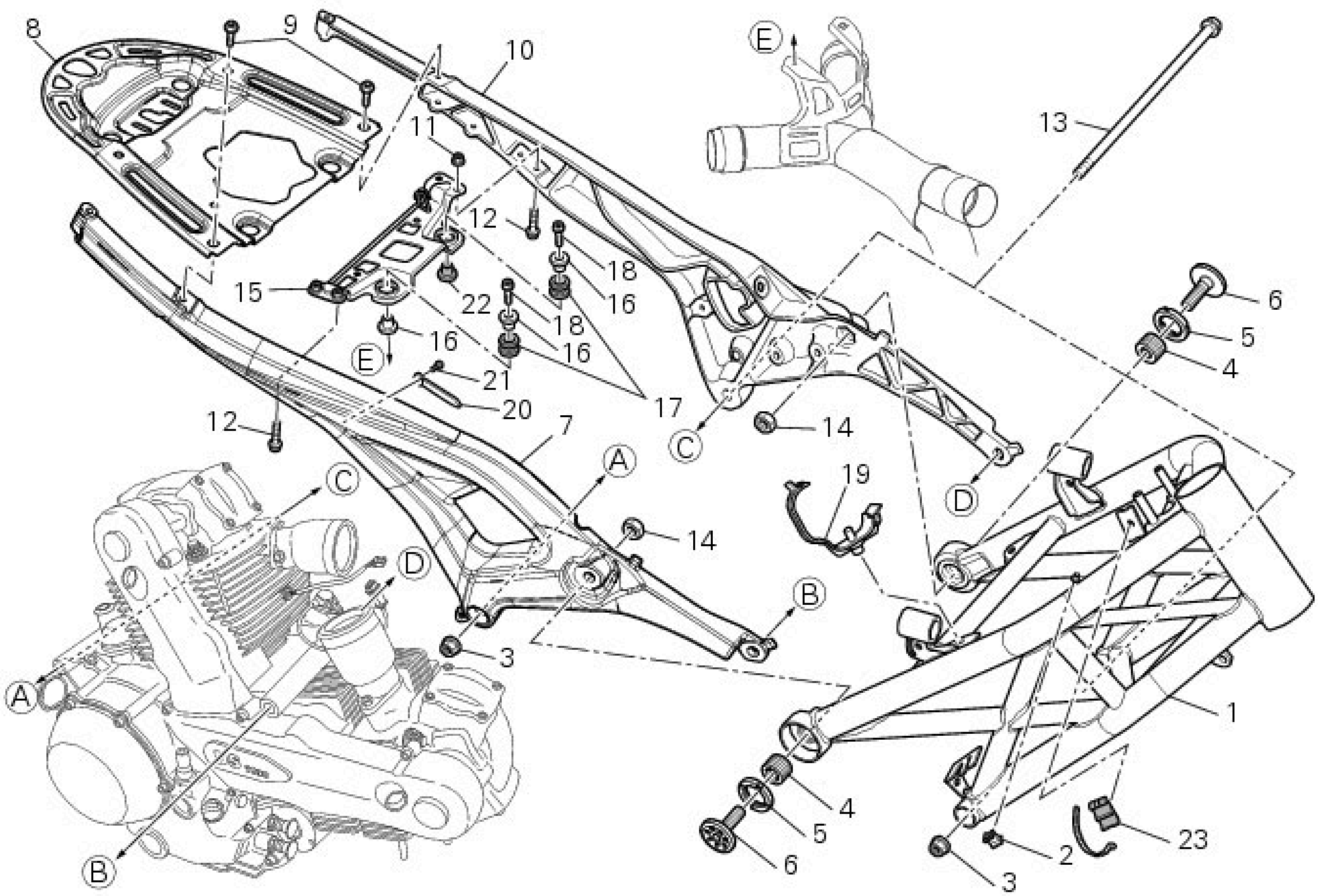
6 -Frame inspection



1 Frame
2 Clamp
3 Nut
4 Clearance adjuster
5 Nut
6 Banjo bolt
7 Rear subframe (right)
8 Plate
9 Screw
10 Rear subframe (left)
11 Nut
12 Screw
13 Pin
14 Special nut
15 Bracket
16 Spacer
17 Vibration damper mount
18 Screw
19 Hose clip
20 Key
21 Screw
22 Spacer
23 Rubber pad

Spare parts catalogue

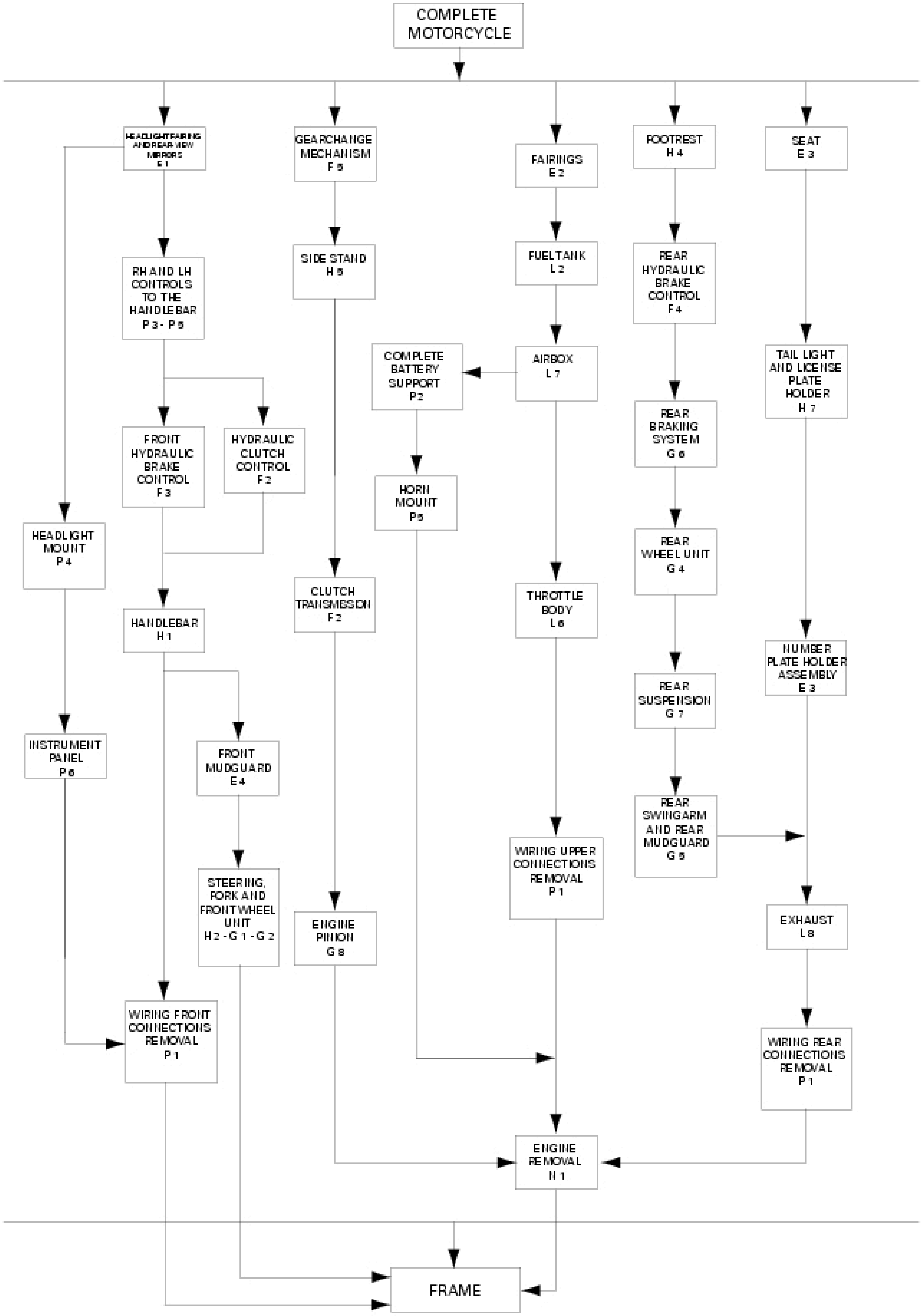
1100 ABS	FRAME
1100 S ABS	FRAME

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

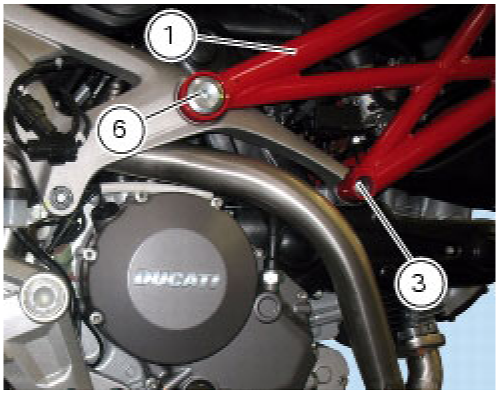
Disassembly of structural components and the frame

Before carrying out dimensional checks on the frame, you must remove all the superstructures fitted, referring to the removal procedures outlined in the sections of this manual.
The flow chart below illustrates the logical sequence in which the parts are to be removed from the motorcycle and gives a reference to the section of the manual in which the removal procedure is described.

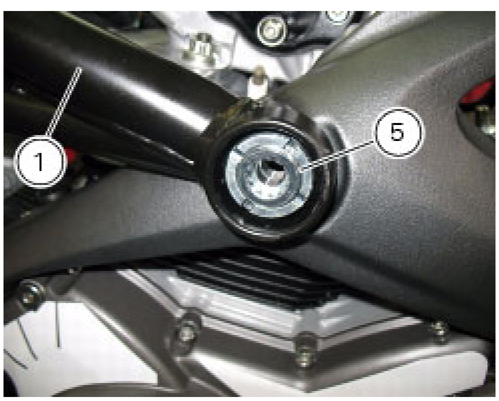

COMPLETE MOTORCYCLE
HEADLIGHT FAIRING AND REAR-VIEW MIRRORS E1
GEARCHANGE MECHANISM F5
FAIRINGS E2
FOOTREST H4
SEAT E3
RH AND LH CONTROLS TO THE HANDLEBAR P3 - P5
SIDE STAND H5
FUEL TANK L2
REAR HYDRAULIC BRAKE CONTROL F4
COMPLETE BATTERY SUPPORT P2
AIRBOX L7
TAIL LIGHT AND LICENSE PLATE HOLDER H7
REAR BRAKING SYSTEM G6
FRONT HYDRAULIC BRAKE CONTROL F3
HYDRAULIC CLUTCH CONTROL F2
HORN MOUNT P5
HEADLIGHT MOUNT P4
REAR WHEEL UNIT G4
CLUTCH TRANSMISSION F2
THROTTLE BODY L6
HANDLEBAR H1
NUMBER PLATE HOLDER ASSEMBLY E3
REAR SUSPENSION G7
INSTRUMENT PANEL P6
FRONT MUDGUARD E4
REAR SWINGARM AND REAR MUDGUARD G5
WIRING UPPER CONNECTIONS REMOVAL P1
STEERING, FORK AND FRONT WHEEL UNIT H2 - G1 - G2
ENGINE PINION G8
EXHAUST L8
WIRING FRONT CONNECTIONS REMOVAL P1
WIRING REAR CONNECTIONS REMOVAL P1
ENGINE REMOVAL N1
FRAME

Removal of the frame

To separate the frame (1) from the rear subframe, unscrew the two special screws (6).
On the left-hand side of the motorcycle, restrain the pin (13) securing the front of the frame to the engine and unscrew the nut (3) on the right-hand side.
Withdraw the front retaining pin (13) from crankcase.

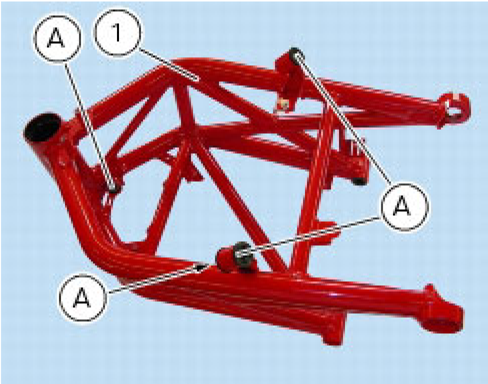


Slacken off the ring nuts (5).
Remove the frame (1) from the rear subframe.

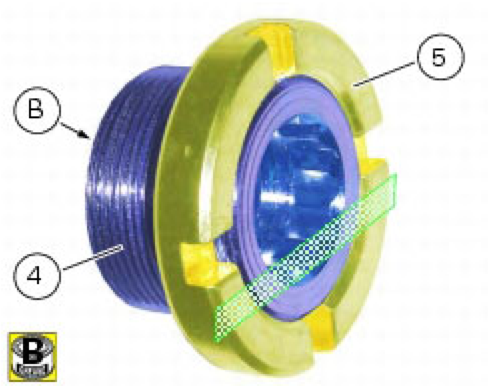


Refitting the frame

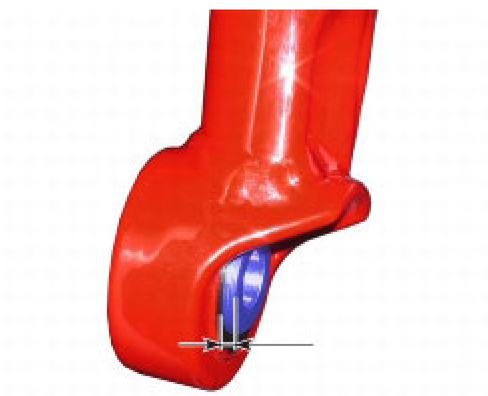
Before refitting, check that the rubber mountings (A) are installed on the frame (1).



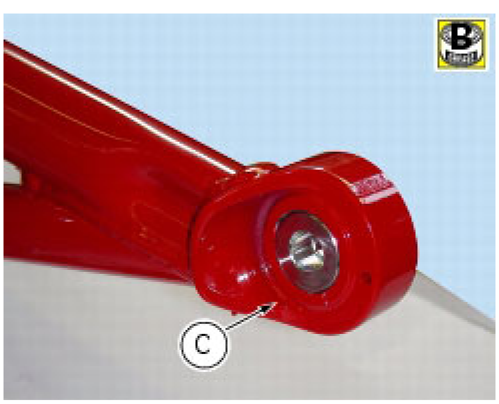
If the ring nuts (5) with the clearance adjusters (4) were removed, proceed as follows.
Grease the threads of the clearance adjusters (4) and the ring nuts (5) taking care not to get any grease on face (B) of the clearance adjuster (4).
Tighten the clearance adjusters (4) onto ring nut (5) until flush with the side featuring flats, as shown below.



Tighten to a snug torque. From the outside, start the clearance adjusters (4) with ring nuts (5) in their thread onto frame threaded bushings (Sect. C 3, Frame torque settings) until they protrude by about **2** mm on the inside.

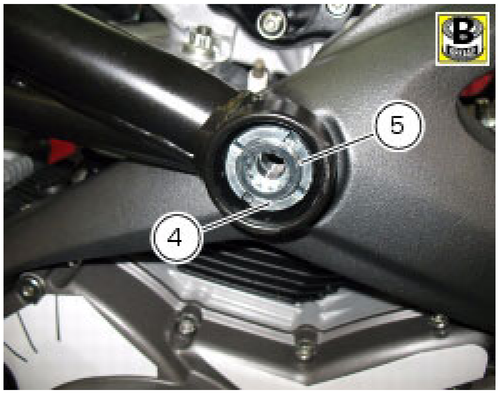


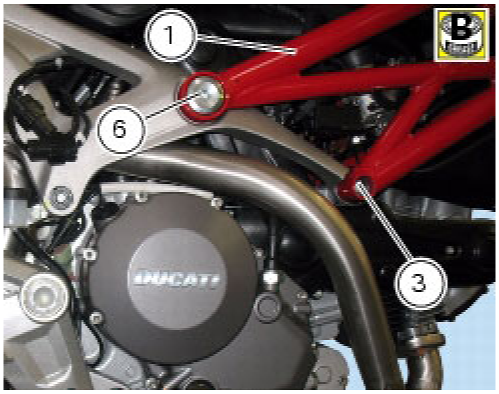
Loosen the clearance adjuster (4) with ring nut (5) until flush with the inner surface of frame bushes (C).

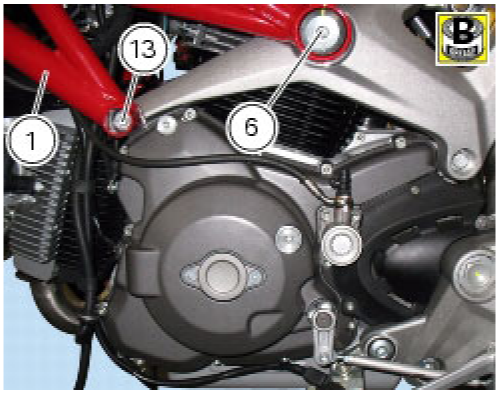


Apply the recommended grease to the threads of the pin (13) and of the nut (3).
Locate the frame (1) on the motorcycle and insert the pin (13) from the right-hand side.
Tighten the nut (3) to the specified torque (Sect. C 3, Frame torque settings) while holding shaft (13).

Align the rear subframe and the frame (1), inserting the special screws (6) but without tightening at this stage.
Tighten the clearance compensator elements (4) to the specified torque (Sect. C 3, Frame torque settings).
Locate service tool no. **88713.3166** on the ring nut (5) and fit the torque wrench to the tool.
While holding the clearance compensator elements (4), tighten ring nuts (5) to the specified torque (Sect. C 3, Frame torque settings).
Lubricate thread and underside of special screws (6), then start them on the frame, and tighten them to the specified torque (Sect. C 3, Frame torque settings).

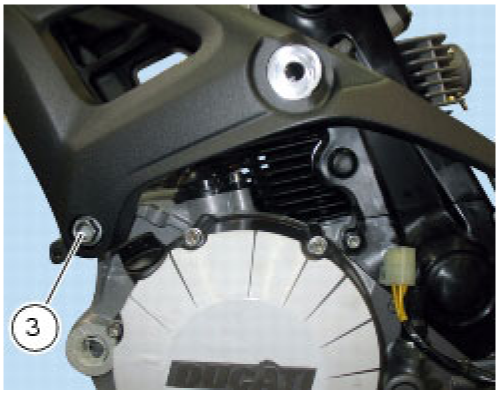






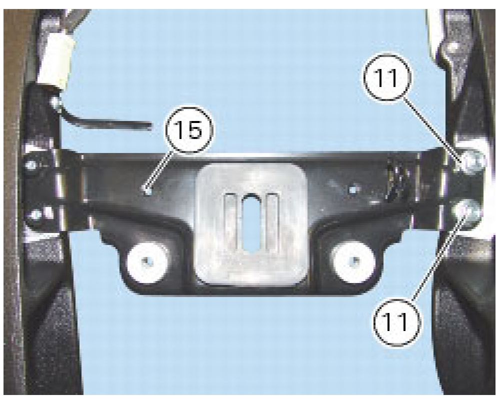
Removal of the rear subframe

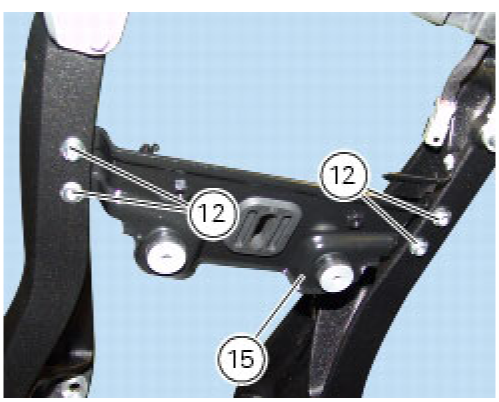
Disconnect all the electrical parts connections from subframe (Sect. P 1, Routing of wiring on frame), and remove the silencers (Sect. L 8, Removal of the exhaust system).
Remove the frame as specified in the previous paragraph "Removal of the frame".
On the left-hand side of the motorcycle, restrain the pin (13) securing the rear of the subframe to the engine and simultaneously unscrew the nut (3) on the right-hand side.
Withdraw the rear retaining pin (13) and remove the complete subframe assembly from the motorcycle.



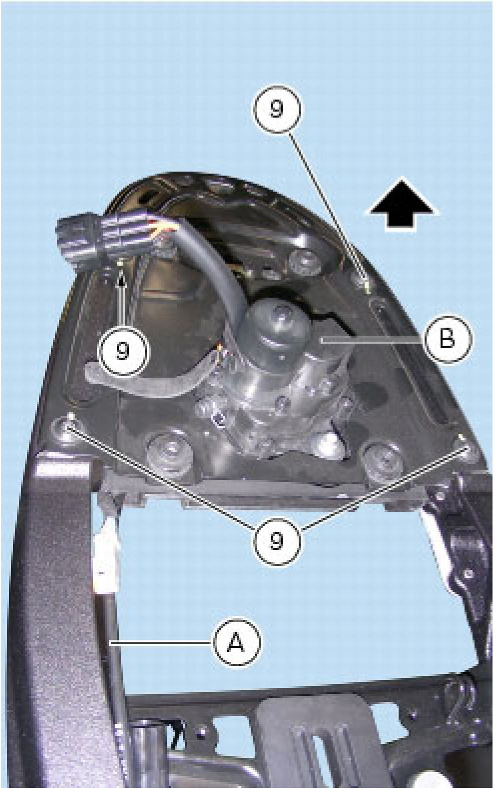
Disassembly of the rear subframe

While holding the nuts (11), present only on the left-hand side of the rear subframe, unscrew the screws (12) and remove the bracket (15).





Remove the cable (A) of exhaust valve motor (B) as explained in Sect. L 8, Removal of the exhaust system.
Unscrew the screws (9) securing the plate (8) and slide out the plate then separate the rear left-hand subframe from the right-hand side one.



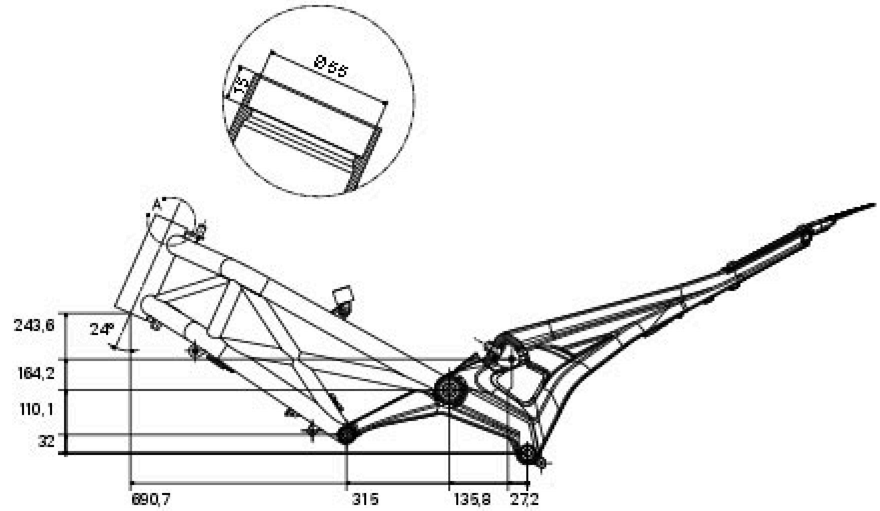
Checking the frame

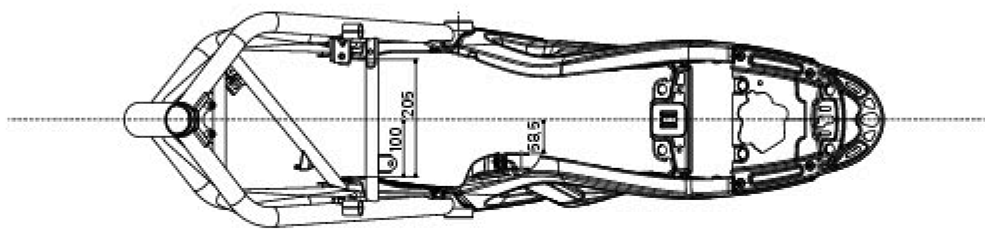
Check the dimensions of the frame against the dimensions shown here to determine whether it needs to be renewed.

Important

Badly damaged frames cannot be repaired, they must be renewed. Any work carried out on the frame can give rise to potential danger, infringing the requirements of EC directives concerning manufacturers' liability and general product safety.

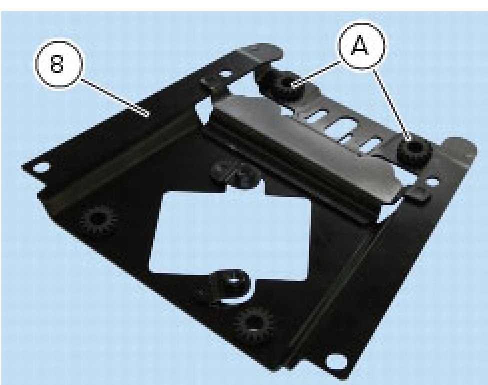
General dimensions (mm)



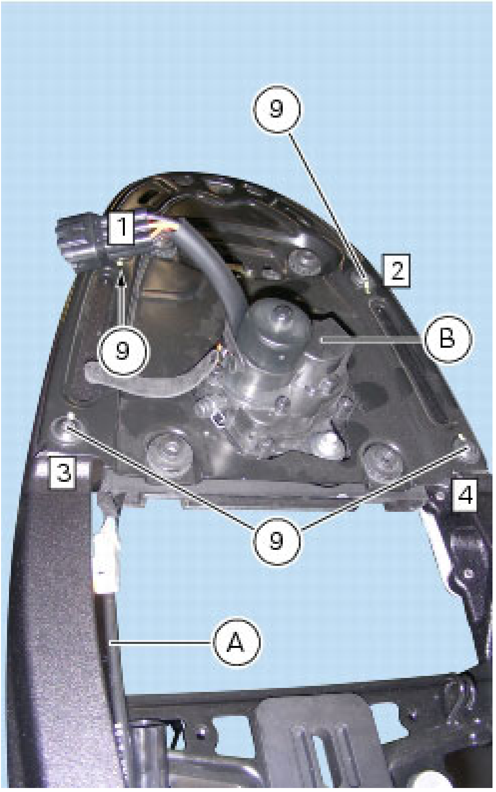


Reassembly of the rear subframe

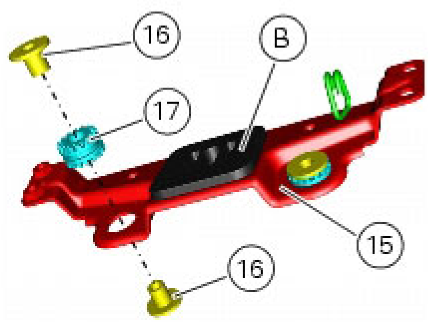
Check that the rubber mountings (A) are installed on the plate (8).



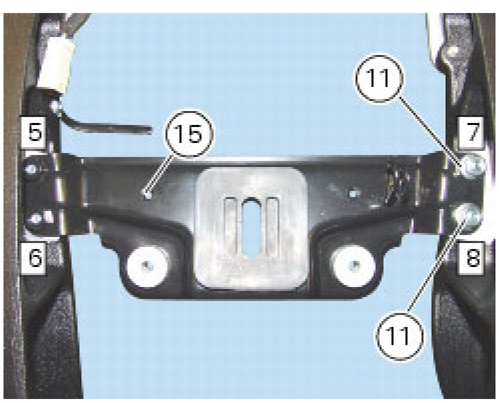
Locate the plate (8) on the LH subframe (10) and on the RH subframe (12) and insert the screws (9).
Tighten the screws (9) to the specified torque (Sect. C 3, Frame torque settings), observing the sequence: 1-2-3-4.
Connect the cable (A) of exhaust valve motor (B) as explained in Sect. L 8, Refitting the exhaust system.

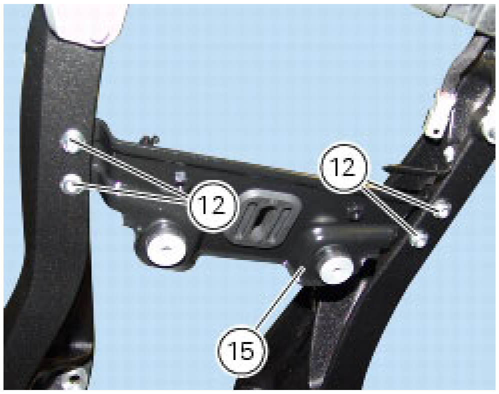


Check that the rubber buffer (B), rubber bushes (17) and spacers (16) are installed on the bracket (15).



Refit the bracket (15) to the rear subframe with screws (12).
While holding the nuts (11), present only on the left-hand side, tighten the screws (12) to the specified torque (Sect. C 3, [Frame torque settings](#)) following the sequence below: 5-6-7-8.





Refit the rear subframe to the crankcase (Sect. N 1, Refitting the engine).
Reconnect all the electrical parts connectors on subframe (Sect. P 1, Routing of wiring on frame), and refit the silencers (Sect. L 8, Removal of the exhaust system).

Refitting the rear subframe

Locate the complete rear subframe assembly on the engine, apply the recommended grease to the threads of the pin (13) and the nut (3).
Tighten the nut (3) to the specified torque (Sect. C 3, Frame torque settings) while holding shaft (13).